

Discrete Math HW 1 Solutions

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1.1 *Proof.* $(x - a)(x - b)(x - c)\dots(x - z) = 0$

1. Observe that 0 times anything equals 0.

2. Observe that $(x - x) = 0$, and that multiplication is commutative.

3. Then

$(x - a)(x - b)(x - c)\dots(x - w)(x - x)(x - y)(x - z) = 0 \cdot (x - a)(x - b)(x - c)\dots(x - w)(x - y)(x - z) = 0$.

4. Therefore $(x - a)(x - b)(x - c)\dots(x - z) = 0$.

□

3.1 Let a and b be integers in the equation $bc = a$.

a) $3|100$ is false. Let $b = 3$ and $a = 100$. There does not exist an integer c such that $3c = 100$.

b) $3|99$ is true. Let $b = 3$ and $a = 99$. There does exist an integer c such that $3c = 99$, namely $c = 33$.

c) $-3|3$ is true. Let $b = -3$ and $a = 3$. There does exist an integer c such that $-3c = 3$, namely $c = -1$.

d) $-5|-5$ is true. Let $b = -5$ and $a = -5$. There does exist an integer c such that $-5c = -5$, namely $c = 1$.

e) $-2|-7$ is false. Let $b = -2$ and $a = -7$. There does not exist an integer c such that $-2c = -7$.

f) $0|4$ is false. Let $b = 0$ and $a = 4$. There does not exist an integer c such that $0c = 4$.

g) $4|0$ is true. Let $b = 4$ and $a = 0$. There does exist an integer c such that $4c = 0$, namely $c = 0$.

h) $0|0$ is true. Let $b = 0$ and $a = 0$. There does exist an integer c such that $0c = 0$, namely $c = 0$.

3.3 Definition 3.5 requires prime numbers to be integers, to be greater than 1, and to have 1 and the number itself as its only positive divisors.

a) $21 = 3 \times 7$ and therefore is not prime.

b) $0 < 1$ and therefore is not prime.

c) π is not an integer and therefore is not prime.

d) $\frac{1}{2}$ is not an integer and therefore is not prime.

e) $-2 < 1$ and therefore is not prime.

f) $-1 < 1$ and therefore is not prime.

3.5 a) Let a be any integer. Let b be a rational number such that $b = \frac{a}{1}$. Then any integer a is equivalent to the rational number b , and therefore every integer is a rational number.

b) Let the rational number $b = \frac{1}{2}$, and observe that b is not an integer. Since there exists a rational number b such that it is not an integer, then not all rational numbers are integers.

3.6 An integer a is called a perfect square provided there exists some integer b such that $a = b^2$.

3.7 A positive number a is called the square root of some number b provided $b = a^2$.

3.9 a) Suppose A , B , and C are points in the plane, and that the function d gives the distance between two points. We say that C is between A and B provided that $d(A, C) + d(C, B) = d(A, B)$.

b) Suppose A , B , and C are points in the plane. We say that they are collinear provided that one of the points is between the other two.

- 3.11 a) A person is called a teenager provided that their age lies between 13 and 19 inclusively.
 b) A person is called a grandmother provided that they are female and that one of their children have a child.
 c) A year is called a leap year provided that it has 366 days with 29 days occurring in February.
 d) A coin is called a dime provided that it is American currency with an inscription of Franklin Roosevelt.
 e) A sequence is called a palindrome provided that it reads the same forwards and backwards.
 f) A word is called a homophone provide that there exists some second word with the same pronunciation but a different meaning.

- 3.13 a) 6
 b) Don't do

- 4.1 a) If an odd and even integer are multiplied together, then their product is even.
 b) If an odd integer is squared, then the square is odd.
 c) If a prime number is squared, then the result is not prime.
 d) If two negative integers are multiplied together, then their product is negative.
 e) If a polygon is a rhombus, then its diagonals are perpendicular.
 f) If two triangles are congruent, then they have the same area.
 g) If a sum is composed of three consecutive integers, then the sum is divisible by three.

- 4.2 I will only bother writing it if it is true, otherwise it may be assumed to be considered false by me. Note: A if and only if B implies all three are true.

- a) If B , then A .
 b) If A , then B .
 c) If A , then B .
 d) If A , then B .
 e) If B , then A .
 f) A and B are contradictions, so none are true.
 g) If A , then B .
 h) A if and only if B .
 i) A if and only if B .
 j) If B , then A .
 k) If B , then A .

- 4.3 Let $A = "x \text{ is a member of the integers}"$, and let $B = "x \text{ is a member of the real numbers}"$.

- a) If A , then B is *true*
 b) If B , then A is *false*

- 4.4 The statements "If a then b " ($a \Rightarrow b$) and "(not a) or b " ($\neg a \vee b$) are logically equivalent because they give the same truth value for every possible combination of *true* and *false*.

Proof.

a	b	$a \Rightarrow b$	$\neg a \vee b$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

□

- 4.7 An equilateral triangle is not a right triangle, and so the antecedent of the Pythagorean Theorem is *false* making the implication of the Theorem still *true*.

4.12 a) The sum of the first n consecutive odd integers is n^2

b) The sum of the first n consecutive odd cubes is $n^2 \cdot (2n^2 - 1)$.

n	1	2	3	4	5
$f(n)$	1	28	153	496	1225
$f(n)/n^2$	1	7	17	31	49
$2n^2$	2	8	18	32	50
$2n^2 - 1$	1	7	17	31	49

c) Note: the problem doesn't state it, but no three lines intersect at a common point. When there are k lines, then adding a new line l adds $k + 1$ new regions. The reason being that the line l is divided into $k + 1$ segments, where each of those segment divides their area into two. Thus adding the $(k + 1)$ th line adds $k + 1$ new regions. Thus no lines has 1 region, the first line adds 1 region, the second adds 2 regions, the third adds 3 regions, etc. Thus for n lines, there are $\frac{n(n+1)}{2} + 1$ regions.